Venkata Lakshmi Parimala Pasupuleti

Forward Deployed AI Engineer · Fraud Detection · Agentic AI · Evaluation Pipelines · Python · Financial Crime · 9yr Production

+1-703-261-5410 | venkatalakshmiparimalap@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Virginia, USA

PROFESSIONAL SUMMARY

Forward deployed AI engineer with 9+ years of production experience, 7 years of hands-on fraud detection, and 2+ years shipping production agentic AI systems. At Entrata, I was the human fraud detection layer for income misrepresentation, fake pay stubs, identity fraud, asset concealment, and household composition fraud across LIHTC/HCV/HUD affordable housing certifications - the same income and identity fraud Sardine's platform fights at scale - then automated 80% of that manual review workload with zero compliance violations. I build and deploy custom AI agents end-to-end: LeaseI is a production 8-agent LangGraph system with conditional routing, HITL guardrails for regulated decisions, and shared memory across agents and sessions. Magizhs was the FDE proof point: sole technical owner, external client, ambiguous requirements to production deployment in weeks. AgentProbe is the evaluation pipeline and AI risk metrics infrastructure - 5-metric quality engine (correctness, groundedness, hallucination rate, latency, token efficiency), LLM-as-Judge scoring, pass@k multi-trial harness, and CI-integrated regression gate - the exact framework needed to trust deployed fraud agents in production. I embed, scope, build, and own; M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

AI Agents & Fraud Detection: Python (9yr), SQL, LangGraph, LangChain, MCP, Multi-Agent Systems, Agent Orchestration, Tool Calling, HITL Guardrails, Intelligent Workflows, Agentic AI, Prompt Engineering, Few-Shot Prompting, Chain-of-Thought, Output Validation, Context Engineering, Anthropic API, OpenAI API, Ollama, Claude Code, Income Fraud Detection, Identity Fraud, Document Intelligence, Financial Crime, Anomaly Detection

Evaluation & Risk Metrics: Evaluation Frameworks, Evaluation Pipelines, AI Risk Metrics, LLM-as-Judge, Hallucination Detection, Groundedness, pass@k Harness, CI Regression Gate, Drift Detection, Observability, Tracing, Feedback Loops, RAG Pipelines, ChromaDB, Vector Databases, Semantic Search, FastAPI, REST APIs, GitHub Actions CI/CD

Data & Infrastructure: Isolation Forest, XGBoost, Scikit-learn, Random Forest, Feature Engineering, Statistical Modeling, PyTorch, AWS (S3, EC2, RDS, Glue), Docker, Microservices, RabbitMQ, PostgreSQL, MySQL, Oracle, Redis, React.js, TypeScript, Linux, Git, Agile/Scrum, PHI/HIPAA Compliance

PROJECTS

Leasel: Production AI Agent System - Fraud-Aware Document Intelligence

Multi-Agent · LangGraph · HITL · Fraud Detection · Production

Python · LangGraph (8-agent) · FastAPI · RAG · SQLite · MCP · Claude Code | github.com/venkatalakshmiparimala/Lease_complaince_rag

Production multi-agent system for fraud-aware eligibility investigation and compliance automation - the same pattern Sardine applies to fincrime workflows. **Agent orchestration:** 8-agent LangGraph StateGraph (Triage → Retrieval → Configuration → Historical → Structured Data → Reasoning → Validation → Escalation) with conditional routing based on confidence scores. **HITL guardrails:** deterministic escalation blocks any automated decision on regulated eligibility paths - LLM reasons, human or deterministic logic decides every final regulated call. **Fraud tracing:** Phase 3 investigation correctly traces false-ineligibility to misconfigured system setting changed by another team - the kind of multi-source fraud pattern that defeats single-system analysis. 93.6% precision, 87.9% F1 across 1,240 test cases.

AgentProbe: AI Risk Metrics & Evaluation Pipeline - Production Fraud Agents

Evaluation Framework · Risk Metrics · CI Gate · LLMOps · Open Source

Python · FastAPI · LangGraph · ChromaDB · MCP · GitHub Actions · Anthropic API · OpenAI API | github.com/venkatalakshmiparimala/AgentProbe

Production evaluation framework and AI risk metrics infrastructure - the tooling needed to trust deployed fraud agents at scale. **Evaluation engine:** 5-metric quality engine (correctness, groundedness, hallucination rate, latency, token efficiency); LLM-as-Judge scoring; pass@k multi-trial harness surfacing variance in non-deterministic outputs. **CI regression gate:** GitHub Actions integration blocks quality drift before deployment - making evaluation a first-class engineering practice. **Platform:** multi-provider LLM routing (Anthropic, OpenAI, Ollama); MCP integration; 15-endpoint FastAPI backend. Open-source, enterprise-deployable. Zero regressions across 1,240+ test cases.

AI on FHIR: Agentic System with Identity Verification & Audit Trail

Full Stack · HIPAA · RBAC · Audit Trail · AWS

Python · Flask · React.js · TypeScript · Docker · AWS · BERT · spaCy · REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack agentic system with LLM-powered tool invocation translating natural-language queries into structured API calls against regulated healthcare data - HIPAA-compliant RBAC with authentication, full data isolation, and complete audit trail enabling every decision to be traced and reviewed. Demonstrates secure agentic system design in regulated domains: identity verification, access controls, and the audit infrastructure fraud prevention requires.

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - Forward Deployed AI Delivery

- Embedded with external client as sole technical owner - translated ambiguous operational requirements into a production agentic AI system (LangGraph, FastAPI, Python) from discovery through architecture, implementation, evaluation pipeline, and deployment in weeks; built HITL guardrails and validation gates for regulated decision paths; delivered observability and feedback loop infrastructure for ongoing operation. Direct FDE delivery pattern: client problem → scoped requirements → production system.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built ML pipelines (PyTorch, scikit-learn; R^2=0.977); applied Isolation Forest anomaly detection across 3 research pipelines surfacing anomalies manual review missed; mentored 600+ students in Python, SQL, and data engineering; delivered dashboards for faculty leadership (40% comprehension improvement).

Entrata - PropTech / Affordable Housing SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **7 years as the human fraud detection layer** - reviewed and flagged income misrepresentation, fake pay stubs, identity fraud, asset concealment (out-of-state and joint accounts), and household composition fraud across LIHTC/HCV/HUD affordable housing certifications; applied Section 42 income and asset limits; cross-referenced household declarations against income documentation under fair-housing constraints requiring uniform, document-driven verification. Entrata published industry research on this problem (fake pay stub fraud up 63% among property managers). Then automated 80% of this manual review workload via Python/PL-SQL pipelines - zero compliance violations over the entire ownership period.
- **Production systems at scale** - owned HUD income-limit import pipeline (180,000 domain records, 100K+ housing units) for 7 years at 99.5% uptime; RabbitMQ event-driven async with retry logic, idempotency guarantees, and dead-letter handling; AWS S3 document migration (100% PHI compliance, zero data loss); Docker microservices redesign (50% scalability improvement); 200+ SQL optimizations (30% performance gain); GitHub Actions CI/CD with automated testing.
- **End-to-end ownership and delivery** - led projects spanning multiple engineers; wrote design docs and architecture decisions; mentored 5+ engineers; 300+ peer code reviews; 20+ full-stack production modules across 120+ Agile sprints; cross-functional delivery with product, compliance, and operations stakeholders.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built income limits import pipeline (1.6M HUD/Tax Credit records across 5 years); property management and compliance modules (Python, PostgreSQL); AWS S3 cloud migration resolving 90% of legacy storage constraints.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017